Memorial Pathology Associates

Integrated Diagnostic Pathology Report

Accession:	MPA-2026-018472	**Patient ID:**	PID-87432
Report date:	2026-04-08	**Age / Sex:**	47 / F

Clinical history / indication

47-year-old woman presenting with new-onset focal motor seizures over six weeks. MRI brain demonstrated a 3.8 cm right frontal lobe mass with patchy enhancement and moderate surrounding edema. No prior cancer history. No family history of CNS tumors. Underwent gross total resection.

Specimen

Right frontal lobe mass, gross total resection (4.1 x 3.2 x 2.6 cm aggregate).

Macroscopic findings

Multiple soft, gray-tan tissue fragments with focal hemorrhage. No grossly necrotic or cystic areas identified.

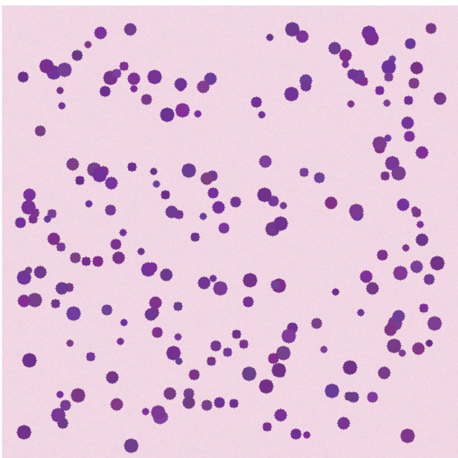
Microscopic / histopathology

Sections demonstrate a moderately to markedly hypercellular astrocytic proliferation infiltrating cerebral cortex and white matter. Tumor cells show moderate nuclear pleomorphism with elongated, irregular nuclei and visible nucleoli. Mitotic figures are identified at up to 4 per 10 high-power fields. Microvascular proliferation and tumor necrosis are NOT identified. No oligodendroglial component is appreciated.

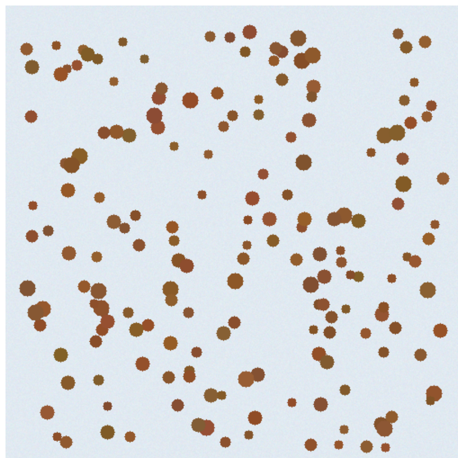
Immunohistochemistry

Marker	Result
GFAP	Positive (diffuse, strong)
IDH1 R132H	Positive (strong cytoplasmic)
ATRX	Loss of nuclear expression in tumor cells
p53	Strong nuclear positivity in >75% of tumor cells
Ki-67 (MIB-1)	Approximately 15% in hotspots
Olig2	Positive
EMA	Negative

Representative images



H&E; ×20 — frontal lobe tumor



IDH1 R132H IHC ×20

Molecular findings

Single-nucleotide variants / indels

Gene	HGVSc	HGVSp	VAF	Classification
IDH1	c.395G>A	p.R132H	0.42	Pathogenic
TP53	c.524G>A	p.R175H	0.38	Pathogenic
ATRX	c.6018del	p.K2007Nfs*4	0.41	Pathogenic

Structural variants / signatures

Kind	Description
chromosomal	1p/19q non-codeleted (intact)

MSI status	MSS
TMB (mut/Mb)	2.4
MGMT promoter	Unmethylated

Pathologist comments

Findings — diffuse astrocytic glioma with IDH1 R132H mutation, ATRX loss, and p53 overexpression. Histologic features (mitotic activity present, no MVP or necrosis) place this at the upper end of grade 3 territory. 1p/19q intact argues against oligodendroglioma. MGMT promoter unmethylated has therapeutic implications (less anticipated benefit from temozolomide). No EGFR amp / no +7/-10 / no TERT promoter mutation detected; not consistent with IDH-wt GBM.

Electronically signed: **Reporting Pathologist, MD, PhD**
2026-04-08